

A Connectome Test of the Fly Hashing Algorithm

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Abstract—Dasgupta, Stevens and Navlakha (2017) described the *Drosophila* olfactory circuit as a locality-sensitive hash. Its glomerulus-to-Kenyon-cell projection was drawn at random because the wiring was unknown. We first rerun their benchmark protocol (SIFT, GloVe, MNIST) with random matrices and recover their result: on MNIST at hash length 4, LSH scores 0.159 (reported 0.160) and the fly hash 0.365, a $2.3\times$ advantage (reported $2.8\times$). We then run the same protocol through the measured projection of the MaleCNS connectome, 51 glomeruli and 1,886 Kenyon cells with every reconstructed connection. The advantage over LSH at short hash lengths survives ($2.1\times$ on MNIST), and the measured wiring scores within -1.9% to +0.3% of uniform rewirings with the same degrees. On mixtures of DoOR odorant profiles it scores -2.06% below them (two-sided $p = 0.020$; 90% interval -3.1% to -0.8%), a deficit that disappears when unmeasured responses are imputed. The headline advantage depends on the budget: the fly network behind it uses up to two orders of magnitude more operations than k -bit LSH, and LSH given equal operations wins on all three benchmarks.

Index Terms—connectomics, locality-sensitive hashing, similarity search, replication

I. INTRODUCTION

Locality-sensitive hashing (LSH) maps items to short codes under which similar items collide with high probability [1], [2]. Dasgupta *et al.* [3] noticed that the fly olfactory circuit computes a related transform with three unusual features: a sparse binary projection, an expansion rather than a reduction of dimension, and winner-take-all. On SIFT, GloVe and MNIST it beat classical LSH by up to about $3\times$. The projection was sampled at random, anchored to biology only by the number of inputs per Kenyon cell, following the account that this wiring is effectively random [4].

That account has since been revised: in a whole-brain electron microscopy volume, projection-neuron input to Kenyon cells is structured [5]. The MaleCNS connectome [6] gives a complete measured projection for a second animal. We ask whether the random matrix was a harmless idealisation, and we do it in the order a replication requires:

- rerun the 2017 benchmark protocol with random matrices and check that it reproduces the published numbers;
- keep that protocol and data, substitute the measured projection, and compare it with uniform rewirings that preserve both degree sequences;
- repeat the comparison on odours, the input the circuit actually receives, with a pre-specified inference protocol;
- compare with LSH at explicit budgets: hash length, storage and operation count.

II. METHODS

A. Measured projection

Let $M \in \{0,1\}^{d \times m}$ with $M_{ji} = 1$ iff glomerulus j connects to Kenyon cell i . We build M from MaleCNS v1.0 using every reconstructed connection, with no synapse-count threshold. Projection neurons of class ALPN are assigned to glomeruli by cell type; multiglomerular, unnamed, thermo- and hygro-sensory types are excluded. The right hemisphere gives $d = 51$ glomeruli and $m = 1,886$ Kenyon cells with, on average, 5.28 distinct glomerular inputs per cell. Because projection neurons are aggregated by glomerulus this is not a claw count; cells have on average 5.59 distinct projection-neuron partners.

B. The 2017 protocol

We follow [3]: 10,000 vectors per dataset (SIFT, $d = 128$ [11]; GloVe, $d = 300$ [12]; MNIST, $d = 784$ [13]); each input centred by its own mean; 1,000 random queries; for each, the true top 2% (200) neighbours by Euclidean distance and the predicted top 2% by Euclidean distance between hashes; mean average precision (mAP), averaged over 50 trials in which matrices and queries change. The fly hash is $y = M^\top x$ followed by the k largest entries set to one. LSH is k Gaussian projections kept as real values, the form compared in [3]; one-bit (sign) LSH scores much lower and is in the supplement. The fly uses $m = 20k$ Kenyon cells, which equates the operation count with LSH under the paper's accounting (d multiplications plus d additions per Gaussian projection, one addition per non-zero of M), or $m = 10d$ as in its headline comparison. Two details are not in the main text and are taken from the FlyLSH reference code [10]: each Kenyon cell samples 10% of the inputs, and the tag is binary. The paper computes mAP by its ref. 19; we use the standard definition (mean precision at the ranks of true neighbours). The list-overlap score of the reference code gives the same ordering and is in the supplement.

C. The measured projection on the 2017 benchmarks

M accepts only $d = 51$ inputs, so each dataset is reduced by PCA to 51 components. Features of images or words have no correspondence to glomeruli, so each trial assigns components to glomeruli by a fresh random permutation. Ground truth is computed on the reduced input, which is what every hash sees. On that input we compare, over 20 trials, the measured M ; 50 null matrices (below); the 2017 construction at the same size, in which every cell samples six glomeruli uniformly; and LSH. Each matrix's mAP is averaged over trials and the measured matrix is ranked among the nulls.

TABLE I

REPLICATION WITH RANDOM MATRICES: MAP, MEAN OF 50 TRIALS. REPORTED IN [3]: MNIST $k=4$ LSH 0.160, FLY (10d) 0.448; SIFT $k=4$ RANDOM SELECTION 0.177, FLY (20k) 0.324.

data	k	LSH	random sel.	fly 20k	fly 10d	10d/LSH
SIFT	4	0.174	0.180	0.283	0.357	$2.1\times$
SIFT	16	0.499	0.500	0.576	0.608	$1.2\times$
GloVe	4	0.055	0.056	0.081	0.093	$1.7\times$
GloVe	16	0.134	0.135	0.150	0.153	$1.1\times$
MNIST	4	0.159	0.165	0.272	0.365	$2.3\times$
MNIST	16	0.516	0.516	0.583	0.622	$1.2\times$

D. Null matrices

The null is uniform over binary matrices with the same row and column sums as M , sampled by curveball trades [7], which converge to the uniform distribution [8]. It keeps how many glomeruli each cell samples and how many cells each glomerulus reaches, and randomises only which glomeruli share a cell. We use 30 trade attempts per cell; glomerulus co-occurrence $Q = \sum_{j<l} (MM^T)_{jl}^2$ stays within three null SDs from 1 sweep on, and retrieval scores at 30 and 100 sweeps agree (supplement).

E. Odour benchmark and inference

Responses are the DoOR 2.0 consensus matrix [9] (commit db323a4), mapped from receptor to glomerulus. DoOR coverage is uneven: 35 glomeruli are measured for 51 or more odorants, eleven for 12 to 24, and five for none. The baseline keeps the 35 (≥ 40 odorants) and odorants measured on at least 12 of them, leaving 172 odorants with 28.7% of entries unmeasured, set to zero; imputations and a 46-glomerulus variant are sensitivity analyses. The restricted projection has $m = 1,838$ cells and mean 3.87 inputs per cell. The benchmark is 4,000 synthetic linear mixtures of two to five DoOR profiles; inputs are normalised to unit mean as in [3]; ground truth is the $\kappa = 10$ nearest items by Euclidean distance on raw responses; Hamming ties are broken by a fixed random order, identical across matrices.

The primary hash size, fixed before this analysis, is $k = 92$ (5% of cells); $k \in \{8, 16, 32, 64\}$ are secondary and Holm-adjusted. The rank of the measured matrix among $B = 200$ nulls gives an exact randomization p -value. The size of the effect across odorant sets is estimated by a two-stage bootstrap that resamples source odorants, regenerates the benchmark and re-draws 20 null matrices per replicate (200 replicates); its coverage is checked by simulation. Equivalence is assessed at a margin of $\pm 5\%$ relative mAP, fixed before this analysis.

III. RESULTS

A. The 2017 result replicates

LSH matches the published baseline (0.159 against 0.160 on MNIST at $k = 4$), and so does random tag selection (0.180 against 0.177 on SIFT). The fly hash is ahead on every dataset at short hash lengths, by $2.3\times$ on MNIST at $k = 4$, and the advantage falls with k (Table I). Our fly scores are 10–20% below the published ones, so the reproduction is close for the

TABLE II

THE MEASURED PROJECTION ON THE 2017 BENCHMARKS (PCA TO 51 INPUTS, 1,886 CELLS): MAP OF MEASURED WIRING, NULL MEAN ($B = 50$), RELATIVE DIFFERENCE, TWO-SIDED p (FLOOR 0.039), THE 2017 CONSTRUCTION AT THE SAME SIZE, AND LSH WITH k PROJECTIONS.

data	k	measured	null	$\Delta\%$	p	2017	LSH
SIFT	4	0.374	0.375	-0.1	0.784	0.407	0.197
SIFT	16	0.567	0.567	+0.1	0.824	0.572	0.573
SIFT	94	0.722	0.728	-0.8	0.039	0.776	0.902
GloVe	4	0.179	0.178	+0.3	0.471	0.207	0.094
GloVe	16	0.312	0.313	-0.3	0.353	0.318	0.316
GloVe	94	0.466	0.475	-1.9	0.039	0.510	0.777
MNIST	4	0.445	0.446	-0.2	0.471	0.470	0.215
MNIST	16	0.672	0.675	-0.4	0.039	0.681	0.654
MNIST	94	0.805	0.810	-0.7	0.039	0.834	0.935

TABLE III

ODOUR BENCHMARK: MEASURED WIRING AGAINST THE NULL ($B = 200$) AND COMPARISON MATRICES (RELATIVE TO THE NULL MEAN, %). *PRIMARY.

k	real	null	$\Delta\%$	p	p_{Holm}	fan	flat	6 unif.	balanced
8	0.1973	0.2034 ± 0.0085	-3.0	0.468	1.000	+3.7	+26.7	+3.2	
16	0.2879	0.2933 ± 0.0062	-1.8	0.438	1.000	+2.7	+9.7	+3.2	
32	0.3531	0.3575 ± 0.0045	-1.2	0.338	1.000	+3.3	+3.0	+3.6	
64	0.3963	0.4001 ± 0.0039	-0.9	0.348	1.000	+3.4	+0.7	+3.4	
92*	0.4063	0.4148 ± 0.0035	-2.1	0.020	–	+3.2	+0.1	+3.2	

baseline and qualitative for the fly hash; the ordering and its dependence on k are the same.

B. Measured wiring on the same benchmarks

With measured wiring the fly hash keeps its advantage over LSH at short hash lengths: on MNIST at $k = 4$ it scores 0.445 against 0.215 ($2.1\times$; Table II). Relative to the null it is within -1.9% to +0.3% at all 18 dataset–size combinations. At the largest sizes it lies below every null draw (6 of 18 combinations at the p floor, unadjusted), the same direction as on odours. The 2017 construction scores higher than both, by up to 20%: the difference between measured and idealised wiring lies in the degree sequences, not in the pairing the null randomises.

C. Measured wiring on odours

The measured pairing is structured: glomerulus co-occurrence Q is +3.7% above the null mean and exceeds all 200 null draws ($p = 0.005$), in agreement with [5]. Its retrieval at the primary size is 0.4063 against 0.4148 ± 0.0035 , a relative difference of -2.06%, below 199 of 200 draws (two-sided $p = 0.020$; Table III, Fig. 1). Secondary sizes give -3.0% to -0.9%, none significant after Holm adjustment. The two-stage 90% interval is -3.1% to -0.8%; equivalence at $\pm 5\%$ holds, at $\pm 2.5\%$ it does not hold. In simulation the interval covered the true value in 100% of 100 datasets at nominal 90%. Across 26 one-factor variations (Table IV) the difference ranges from -3.9% to +0.8%. The exception is missing data: imputing unmeasured responses gives -0.7% to +0.6% ($p \geq 0.439$), and using all 46 glomeruli with data gives no difference.

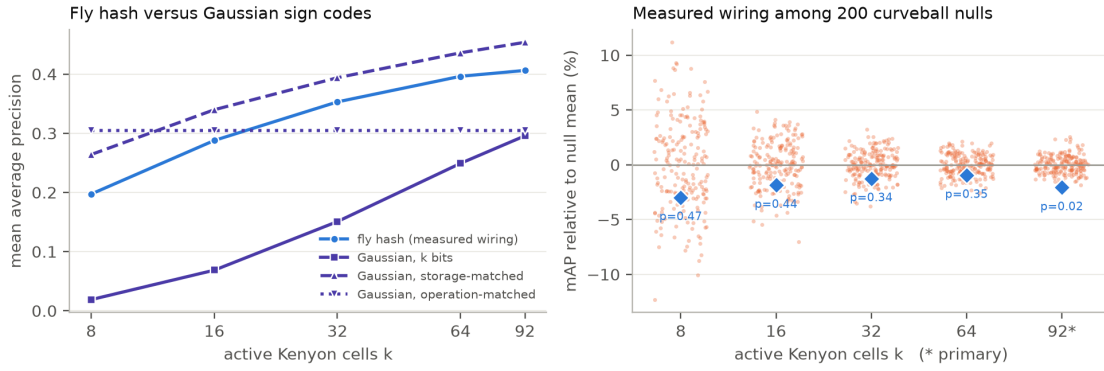


Fig. 1. Odour benchmark. **Left:** mAP of the fly hash on measured wiring and of Gaussian sign codes with k bits, at matched storage and at matched operations (102 bits; mean of 20 seeds). **Right:** each orange point is one of 200 uniform fixed-margin rewirings, blue diamonds the measured wiring, both relative to the null mean.

TABLE IV

ODOUR BENCHMARK, PRIMARY SIZE, UNDER ONE-FACTOR VARIATIONS ($B = 40$ NULLS EACH): RELATIVE DIFFERENCE (%) AND TWO-SIDED p .

variation	$\Delta\%$	p
Baseline	-2.2	0.049
Left hemisphere	-1.1	0.244
≥ 5 -synapse threshold	-2.3	0.098
Missing data (3 imputations)	-0.7 to +0.6	0.439–0.780
Coverage thresholds (2)	-1.7 to -1.4	0.049–0.341
All 46 glomeruli with data, imputed	-0.0	1.000
Mixture seeds (4)	-2.4 to -1.6	0.049–0.244
Mixture size 2 / 2–10	-2.0 to -0.9	0.146–0.195
Measured odorants only	-3.9	0.049
Normalised / angular metric	-2.4 to -1.7	0.049
Tie rules (index, include, 2 seeds)	-2.4 to -2.1	0.049
Synapse-weighted	-1.1	0.488
Synthetic inputs (3)	-3.1 to +0.8	0.049–0.244

TABLE V

ODOUR BENCHMARK: MAP OF THE FLY HASH AND OF GAUSSIAN SIGN CODES (BITS IN PARENTHESES).

k	fly	Gauss. k bits	storage-matched	operation-matched
8	0.197	0.019	0.264 (72)	0.305 (102)
16	0.288	0.069	0.340 (130)	0.305 (102)
32	0.353	0.151	0.394 (229)	0.305 (102)
64	0.396	0.249	0.436 (397)	0.305 (102)
92	0.406	0.296	0.454 (523)	0.305 (102)

D. The advantage depends on the budget

The 2017 comparison sets the fly’s k active cells against k LSH projections. That matches operations only at $m = 20k$. The headline network has $m = 10d$ cells sampling $d/10$ inputs each, about d^2 additions, against $2kd$ operations for LSH: $d/2k$ times more, $98\times$ on MNIST at $k = 4$. LSH given that many operations (390 projections on MNIST) reaches mAP 0.67–0.97 across the three datasets, above every fly score (at most 0.73). The same holds for the measured projection: LSH with the operations of M (98 projections) reaches 0.78–0.94, above the measured fly at every k .

On odours ($d = 35$) the fly hash beats sign codes at k bits ($1.37\times$ at the primary size), trails at matched storage ($0.90\times$), and at matched operations trails for small k and leads from

$k = 32$ ($1.33\times$ at the primary size; Table V). The advantage at k bits is real and replicates; it is an advantage per active cell, not per operation or per stored bit.

IV. DISCUSSION

The random matrix of [3] was a harmless idealisation for this task. Put in its place, the measured projection keeps the fly hash’s advantage over LSH at short hash lengths, and retrieves within two to three percent of random rewirings with the same degrees, on image and word benchmarks as on odours. Where it differs, it is slightly worse, in the direction reported for odour discrimination in [5]. The idealised construction does somewhat better than either because its degrees are uniform; the measured degree sequences, including the skewed fan-out, cost more than the pairing.

Two cautions apply. The odour deficit depends on treating unmeasured DoOR entries as zero and is not detectable under any imputation. On the image and word benchmarks the input has been compressed to 51 dimensions and assigned to glomeruli arbitrarily, so those results test the wiring’s statistics, not its odour-specific pairing. That the measured structure serves other ends, such as innate valence [5] or novelty detection [14], is not tested here.

The larger qualification concerns the benchmark claim itself. Its headline advantage is per active cell. Counted in operations, as [3] count them, the headline network is far more expensive than the LSH it is compared with, and on those benchmarks equal operations reverse the result. At the circuit’s own input size the balance is closer: on odours the fly hash leads sign codes of equal operation count from $k = 32$.

V. LIMITATIONS

One animal, one hemisphere as the primary analysis, one task family. Our fly scores on the 2017 benchmarks are below the published values, and some protocol details were taken from reference code. The odour benchmark is generated from partially observed profiles with linear mixing. The model is binary and rate-free and ignores inhibition beyond winner-take-all.

VI. CONCLUSION

The 2017 fly hashing result replicates with random matrices and survives the substitution of measured wiring: the measured MaleCNS projection performs like a degree-preserving random one, slightly worse at larger hash sizes. Its advantage over LSH holds per active cell and not per operation. Code, data checksums, results and a supplement are at <https://github.com/ssenge/FlyHash-Connectome>.

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